Dongho Kang

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RESEARCH INTERESTS

The goal of my research is to broaden the skill set of humanoid robots and enhance their ability to generate natural and versatile behaviors by integrating advanced control strategies with data-driven character animation techniques.

PROFESSIONAL AFFILIATIONS & ACTIVITIES

RAI Institute, Zurich, Switzerland

- Research Scientist — Jun 2025 – Present
  - Led research on data-driven control frameworks to achieve for natural and steerable humanoid whole-body behaviors.
  - Developed robust perceptive locomotion control pipelines for legged systems.

NVIDIA, Zurich, Switzerland

- Deep Learning Intern — Jun 2018 – Dec 2018
  - Researched deep learning algorithms for super-resolution and anti-aliasing within the DLSS pipeline.

CNP Technology Inc., Seoul, South Korea

- CAD Engineer — Dec 2011 – Mar 2014
  - CAD-designed enclosures and managed end-to-end manufacturing processes for industrial and embedded PCs.

EDUCATION

ETH Zürich, Zurich, Switzerland

- Doctor of Science in Computer Science — Apr 2020 – Aug 2025
  - Main advisor: Prof. Dr. Stelian Coros
  - Second advisor: Prof. Dr. Marco Hutter
  - Thesis: Animal Motion Imitation for Adaptive and Lifelike Locomotion Control of Legged Robots
- Master of Science in Mechanical Engineering — Sep 2016 – Aug 2019
  - Advisor: Prof. Dr. Marco Hutter
  - Thesis: End-to-End Collision Avoidance from Depth Input with Memory-based Deep RL

Seoul National University, Seoul, South Korea

- Bachelor of Science in Mechanical Engineering & Computer Science — Mar 2009 – Aug 2016
  - Advisor: Prof. Dr. Dongjun Lee

PUBLICATIONS

JOURNALS

[1] Mike Zhang, Dongho Kang, Kevin Bergamin (co-first authors); advised by Alfred Rizzi, Jessica Hodgins, Scott Kuindersma, Yeuhi Abe, Sylvain Bertrand, and Farbod Farshidian, "HuMBLE: Human Motion-Driven Behavior Learning for Embodied Locomotion," 2026 (under review)

[2] Matthias Heyrman, Chenhao Li, Victor Klemm, Dongho Kang, Stelian Coros, and Marco Hutter, "Multi-Domain Motion Embedding: Expressive Real-Time Mimicry for Legged Robots," in *The International Journal of Robotics Research (IJRR)*, 2026 (accepted.)

[3] Lukas Molnar, Jin Cheng, Gabriele Fadini, Dongho Kang, Fatemeh Zargarbashi, and Stelian Coros, "Whole-body Inverse Dynamics MPC for Legged Loco-manipulation," in *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[4] Jin Cheng, Dongho Kang, Gabriele Fadini, Guanya Shi, and Stelian Coros, "RAMBO: RL-augmented Model-based Optimal Control for Whole-body Loco-manipulation," in *IEEE Robotics and Automation Letters (RA-L)*, Sep 2025.

[5] Taerim Yoon, Dongho Kang, Seungmin Kim, Jin Cheng, Minsung Ahn, Stelian Coros, and Sungjoon Choi, "Spatio-Temporal Motion Retargeting," in *IEEE Transactions on Robotics (T-RO)*, Aug 2025.

[6] Dongho Kang, Jin Cheng, Miguel Zamora, Fatemeh Zargarbashi, and Stelian Coros, "RL + Model-based Control: Using On-demand Optimal Control to Learn Versatile Legged Locomotion," in *IEEE Robotics and Automation Letters (RA-L)*, Oct 2023.

CONFERENCES

[1] Taerim Yoon, Dongho Kang, Kisang Park, Junha Cha, Stelian Coros, Sungjoon Choi, "Learning Unions of Convex Sets via Invertible Latent Decomposition for Path Planning," in *Conference on Robot Learning (CoRL)*, 2026.

[2] Taerim Yoon, Dongho Kang, Jin Cheng, Fatemeh Zargarbashi, Yijiang Huang, Minsung Ahn, Stelian Coros, and Sungjoon Choi, "Teaching Robots Like Dogs: Learning Agile Navigation from Luring, Gesture, and Speech," in *Conference on Robot Learning (CoRL)*, 2026.

[3] Yarden As, Chengrui Qu, Benjamin Unger, Dongho Kang, Max van der Hart, Laixi Shi, Stelian Coros, Adam Wierman, Andreas Krause, "SPiDR: A Simple Approach for Zero-Shot Safety in Sim-to-Real Transfer," in *Neural Information Processing Systems (NeurIPS)*, 2025.

[4] Fatemeh Zargarbashi, Jin Cheng, Dongho Kang, Robert Sumner, and Stelian Coros, "RobotKeyframing: Learning Locomotion with High-Level Objectives via Mixture of Dense and Sparse Rewards," in *Conference on Robot Learning (CoRL)*, 2024.

[5] Adrian Hartmann, Dongho Kang, Fatemeh Zargarbashi, Miguel Angel Zamora Mora, and Stelian Coros, "Deep Compliant Control for Legged Robots," in *International Conference on Robotics and Automation (ICRA)*, 2024.

[6] Daniel Widmer, Dongho Kang (co-first authors), Bhavya Sukhija, Jonas Hübotter, Andreas Krause, and Stelian Coros, "Tuning Legged Locomotion Controllers via Safe Bayesian Optimization," in *Conference on Robot Learning (CoRL)*, 2023.

[7] Dongho Kang, Flavio De Vincenti, Naomi C. Adam, and Stelian Coros, "Animal Motions on Legged Robots Using Nonlinear Model Predictive Control," in *International Conference on Intelligent Robots and Systems (IROS)*, 2022.

[8] Dongho Kang, Simon Zimmermann, and Stelian Coros, "Animal Gaits on Quadrupedal Robots using Motion Matching and Model-Based Control," in *International Conference on Intelligent Robots and Systems (IROS)*, 2021.

[9] Flavio De Vincenti, Dongho Kang, and Stelian Coros, "Control-Aware Design Optimization for Bio-Inspired Quadruped Robots," in *International Conference on Intelligent Robots and Systems (IROS)*, 2021.

[10] Changu Kim, Hyunsoo Yang, Dongho Kang and Dongjun Lee, "2-D Cooperative Localization with Omni-Directional Mobile Robots," in *International Conference on Ubiquitous Robots and Ambient Intelligence*, 2015.

THESIS

[1] Dongho Kang, "Animal Motion Imitation For Adaptive and Lifelike Control of Legged Robots," Doctoral thesis, Department of Computer Science, ETH Zürich, 2025.

[2] Dongho Kang, "End-to-End Collision Avoidance from Depth Input with Memory-based Deep RL," Master's thesis, Department of Mechanical and Process Engineering, ETH Zürich, Aug 2019.

INVITED TALK

- Institute of Robotics and Mechatronics, German Aerospace Center (DLR) — Jul 2026
  Oberpfaffenhofen-Weßling, Germany
- Illinois Robotics Group, University of Illinois at Urbana-Champaign — Mar 2026
  Champagne-Urbana, Illinois, United States (online)
- Laboratory for Intelligent Decision and Autonomous Robots, Georgia Tech — Feb 2026
  Atlanta, Georgia, United States
- Department of Artificial Intelligence, Korea University — Jan 2026
  Seoul, South Korea
- Department of Computer Science and Engineering, Seoul National University — Jan 2026
  Seoul, South Korea
- Machine Perception for Human Understanding, AI+X Summit 2025 — Oct 2025
  Zurich, Switzerland
- Department of Mechanical Engineering, Seoul National University — Oct 2025
  Seoul, South Korea (online)

- Artificial Intelligence Graduate School, UNIST — Oct 2025
  Ulsan, South Korea (online)
- Biomimetic Robotics Lab, Massachusetts Institute of Technology — Aug 2024
  Cambridge, Massachusetts, United States (online)
- Johou Systems Kougaku Laboratory, University of Tokyo — May 2024
  Tokyo, Japan
- Suzumori Laboratory, Tokyo Institute of Technology — May 2024
  Tokyo, Japan

AWARDS & SCHOLARSHIPS

- Birkigt Scholarship, ETH Zürich — Feb 2018
  Stipendiary scholarship for international master student.
- Eminence Scholarship, Seoul National University — Aug 2014
  Full-tuition scholarship for one academic semester for outstanding academic performance.
- Development Fund Scholarship, Seoul National University — Feb 2010
  Full-tuition scholarship for one academic year for outstanding academic performance.

TEACHING EXPERIENCE

ETH Zürich, Zurich, Switzerland

- Teaching Assistant, Stochastics and ML (A. Streich, C. Cotrini, F. Friedrich) — Spring 2025
- Teaching Assistant, Introduction to Machine Learning (F. Perez-Cruz, F. Yang) — 2024
- Teaching Assistant, Computer Science (M. Fischer, F. Friedrich) — 2023
- Teaching Assistant, Digital Humans (S. Coros, Siyu Tang) — 2023
- Teaching Assistant, Linear Algebra (Ö. Imamoglu, O. Sorkine-Hornung) — 2022
- Teaching Assistant, Computational Models of Motion (S. Coros, B. Thomaszewski) — 2021 – 2022
- Teaching Assistant, Visual Computing (S. Coros, M. Pollefeys) — 2020 – 2021

Seoul National University, Seoul, South Korea

- Mentor, SNU Samsung Convergence Software Course Program — 2015
- Teaching Assistant, MAE 446.204A: Dynamics — 2014
- Teaching Assistant, PA 034.013: Basic Physics 2 — 2011

TECHNICAL SKILLS

Programming and Software

C/C++, Python, Matlab/Octave, Unix/Linux, Tensorflow, Pytorch, ROS, Open Dynamics Engine, IsaacSim

Experience with Robots

UnitreeRobotics AlienGo/A1/Go1/Go2/B2/G1, ANYbotics ANYmal

SERVICES

Reviewer

T-RO, TIE, RA-L, ICRA, IROS, RSS, CoRL, Humanoids, BioRob, Eurographics

LANGUAGES

- Korean: Native language.
- English: Fluent.

REFERENCES

- **Prof. Dr. Stelian Coros**
  Associate Professor, ETH Zürich
  scoros@inf.ethz.ch
- **Prof. Dr. Marco Hutter**
  Professor, ETH Zürich / Senior Director, RAI Institute
  mahutter@ethz.ch
- **Prof. Dr. Dongjun Lee**
  Professor, Seoul National University
  djlee@snu.ac.kr